

Attention and Saliency Map Extraction in Visual AI Models: A Comprehensive Review

Introduction

Deep learning models for vision (e.g. image classifiers and detectors) have achieved remarkable performance, but their decision processes remain largely opaque. **Explainable AI (XAI)** techniques aim to make these “black-box” vision models more interpretable by highlighting where and what the model is “looking at” in an input image ¹ ². In particular, **saliency maps** (also called visual attribution maps or attention maps) provide a heatmap over an image, indicating which regions most influenced the model’s prediction ². This review surveys the state-of-the-art (SOTA) methodologies for extracting such attention/saliency maps across different architectures – from traditional Convolutional Neural Networks (CNNs) to modern Vision Transformers (ViTs) and beyond – and discusses current progress in understanding these maps in the context of interpretability. We cover standard methods for generating saliency maps, how these methods function and what they reveal, the impact of model size and computational efficiency on attention mechanisms, and how model visual attention compares to human visual attention.

Saliency Map Extraction Methods for CNNs

In early XAI research on computer vision, efforts focused on CNNs by attributing each pixel’s contribution to the model’s output ³ ⁴. Over the years, a rich toolbox of **post-hoc saliency methods** has been developed for CNNs, which can be grouped into a few broad categories:

- **Gradient-Based Methods:** These use the network’s gradients to estimate each input pixel’s influence on the output. Simonyan et al. (2014) first introduced the idea of a **saliency map as the gradient magnitude** of the class score with respect to input pixels, producing a coarse highlight of important image regions ². Variants quickly improved upon this: **Guided Backpropagation** (Springenberg et al. 2015) modifies the backpropagation rule to propagate only positive gradients, yielding sharper, edge-focused maps. **Integrated Gradients (IG)** (Sundararajan et al. 2017) accumulates gradients along a path from a baseline (e.g. a black image) to the input, addressing gradient saturation and improving faithfulness by satisfying certain axioms ². **SmoothGrad** (Smilkov et al. 2017) averages gradients over many noisy perturbations of the input to reduce visual noise ². These gradient-based techniques highlight which pixels (or regions) the model’s prediction is most sensitive to, often revealing object outlines or distinctive features. However, raw gradients can be noisy; thus methods like SmoothGrad and IG aim to produce more coherent, robust attributions ⁵.
- **Class Activation Mapping (CAM) Techniques:** Class Activation Maps were originally introduced for CNNs with a global average pooling layer by Zhou et al. (2016), allowing coarse localization of class-specific regions by weighting the final convolutional feature maps with the class prediction weights. **Grad-CAM** (Selvaraju et al. 2017) generalized this idea to any CNN architecture by using **gradient signals** flowing into a chosen convolutional layer to compute importance weights for that layer’s feature maps ⁶ ⁷. The weighted sum of feature maps (passed through ReLU) yields a heatmap that is class-specific yet retains the spatial layout ⁸. Grad-CAM became a

standard approach due to its broad applicability and one-pass efficiency (one forward and one backward pass) ⁹ ¹⁰. This heatmap, upsampled to input resolution and often overlaid on the image, indicates **which image regions “activated” the CNN for the target class** ¹¹ ¹². Extensions of Grad-CAM have improved its resolution and faithfulness: **Grad-CAM++** (Chattopadhyay et al. 2018) incorporates higher-order gradient terms to better handle multiple objects or instances, yielding more refined localization ¹³. **Guided Grad-CAM** combines Grad-CAM’s coarse heatmap with guided backprop’s fine detail, producing high-resolution, class-specific visualizations ¹⁴ ¹⁵. Other variants like **Smooth Grad-CAM++** average results over noisy inputs for stability ¹⁶, and **XGrad-CAM** (Fu et al. 2020) apply principled axioms (sensitivity and conservation) to adjust the weighting, improving faithfulness in theory ¹⁷. All these CAM-based methods provide an intuitive localization of “where” the network focuses for a prediction ¹⁸ – often highlighting the object or its discriminative parts (like a dog’s face for class dog), thus making the model’s reasoning more interpretable.

- **Perturbation and Occlusion Methods:** Instead of using gradients, perturbation-based methods evaluate the model’s output changes when parts of the input are masked or altered. A classic example is the **Occlusion test** (Zeiler & Fergus 2014), which involves sliding a gray or blurred patch over the image and recording the drop in prediction score to map out important regions. More recent methods like **RISE** (Petsiuk et al. 2018) sample many random masks on the image and learn pixel importances from how occluding different regions affects the output ¹⁹. Perturbation approaches are model-agnostic and often very faithful (directly testing the effect of each region on the prediction) ²⁰. However, they tend to be **computationally expensive** – requiring hundreds or thousands of forward passes with masked inputs – which can be prohibitive for large models or real-time use. Despite the cost, these methods can uncover if a model relies on spurious background cues by showing, for example, that occluding the sky drastically lowers “bird” classification confidence (implying the model learned a context bias). They complement gradient-based methods by **directly verifying causal importance**, albeit at the cost of efficiency.
- **Decomposition and LRP-Based Methods:** Another family of saliency techniques stems from Layer-Wise Relevance Propagation (LRP) and related Deep Taylor Decomposition (Bach et al. 2015; Montavon et al. 2017). These methods propagate the prediction “backwards” through the network using tailored rules at each layer to distribute the output score onto the input pixels ³ ²¹. LRP ensures that the relevances at the input sum up to the model’s output score, thus providing a **conservation property** for interpretability. For CNNs, LRP often highlights similar regions as Grad-CAM but with a more theoretically grounded approach (no gradients needed, instead using the network’s actual activations and weights). Variants like **DeepLIFT** (Shrikumar et al. 2017) and **Shapley value-based methods (e.g. Deep SHAP)** also fall in this category, attributing differences from a baseline input to the output. These methods can capture both positive and negative contributions (what features support vs. oppose a prediction) depending on the propagation rules. While **LRP and decomposition methods can yield fine-grained attribution maps**, they require careful handling of different layer types (ReLU, pooling, etc.) and can sometimes produce artifacts or require regularization to avoid concentrating too much relevance on single pixels ²². Nonetheless, they remain an important part of the SOTA toolkit, and recent work has extended LRP specifically to modern architectures (as we’ll discuss for Transformers).
- **Other Approaches:** Beyond the above, there are general model-agnostic explainers like **LIME** and **SHAP** which can be applied to images as well (e.g., by treating superpixels as features). For instance, LIME (Ribeiro et al. 2016) learns a local surrogate linear model by perturbing superpixels, and can output a map of superpixel importances. However, such methods are less commonly used in vision compared to the deep-learning-specific saliency methods, because they often

produce coarser explanations and have high computational cost. Additionally, some newer research has explored intrinsically interpretable vision models (like prototype networks or networks with attention modules) – while those are beyond our scope, it’s worth noting that architectures with built-in attention mechanisms can sometimes output attention maps as part of their operation (e.g. the attention masks in YOLO’s attention modules). Still, even these require careful interpretation and often one still applies the above post-hoc techniques to them for consistency.

Standard CNN Saliency Map Behavior: The aforementioned methods typically produce heatmaps that highlight the parts of the image most **influential for the model’s decision** ². For example, if a CNN classifies an image as “tiger cat”, a saliency map (Grad-CAM, LRP, etc.) will usually light up the cat’s face and stripes, indicating the model focused on those features (and not the background) to identify the cat ¹¹ ²³. This is extremely useful: it provides a sanity check that the model is attending to expected areas (e.g. the animal itself, not irrelevant scenery) ²⁴ ²⁵. It can also reveal failure modes – for instance, a medical CNN might erroneously highlight an image artifact instead of the actual pathology, warning users that the prediction might be unreliable. Thus, saliency maps for CNNs serve as a primary tool to **interpret model reasoning and build user trust** ²⁶ ²⁷.

Visual Attention in Vision Transformers and Advanced Architectures

The rise of **Vision Transformers (ViTs)** and other attention-based architectures has introduced new opportunities and challenges for interpretability. Vision Transformers do not rely on localized convolutional features; instead, they process images as a sequence of patches and use self-attention mechanisms to weigh relationships between patches ²⁸ ²⁹. Intriguingly, the self-attention **weights** in a transformer are themselves a form of attention map: they explicitly indicate how much each patch attends to every other patch (or to a special [CLS] token) at each layer ²⁹. This inherently structured attention is often touted as making transformers “more interpretable” than CNNs, at least structurally ³⁰ ³¹. However, using raw attention weights as explanations has proven non-trivial – leading to an evolving set of methods to extract meaningful saliency from ViTs and other attention networks.

Direct Attention Visualization: A straightforward approach to interpret a ViT is to visualize the attention matrices. For example, one can take the attention weights from the last layer (or an average across heads/layers) and visualize the attention paid to each image patch by the [CLS] token (which is used for classification) ³² ³³. This produces a coarse attention map over the image, often highlighting roughly the object of interest. The original ViT paper itself noted that one can multiply the attention matrices across all layers (with appropriate handling of residual connections) to compute a single overall attention map from input patches to the output token – a technique later dubbed **Attention Rollout** ³³ ³⁴. **Attention Rollout** (Abnar & Zuidema 2020) assumes that attention can be propagated linearly through layers: by multiplying the $N \times N$ attention matrix of each layer (after adding an identity to account for skip connections), one obtains an $N \times N$ matrix representing the overall influence of patch j on patch i across the network ³⁵ ³⁶. By looking at the row corresponding to the [CLS] token, one can see which patches most strongly contribute to the classification ³⁷. This rollout method is simple and fast (just matrix multiplications) and indeed was **one of the first tools to visualize ViT attention** ³⁸ ³⁹. It often shows that the ViT concentrates on the object region by the final layers (effectively segmenting it out) ⁴⁰. However, naive attention aggregation has limitations. Researchers observed that averaging attention across heads or layers can “blur” important patterns and dilute the semantics of deeper layers ⁴¹. Moreover, **attention weights are not class-specific** – they indicate what parts of the image influence all outputs, not why a particular class was chosen ⁴². As a result, raw attention maps may miss fine-grained details or fail to distinguish between different prediction classes.

Augmented or Filtered Attention Maps: To get class-discriminative explanations from transformers, later methods injected additional information (like gradients) into the attention. One such approach is **Attention Flow / Attention Rollout with gradients**. An example is the **Transformer Multi-Modal Explainability (TMME)** method ⁴³, which extends rollout by multiplying each attention head’s weights by the gradient of the output w.r.t that attention weight (essentially a sensitivity analysis) ⁴³. Similarly, the **SAW** method (“Self-Attention Weighting” , Mallick et al. 2022) scales each layer’s attention matrix by the gradient of the output with respect to that attention, before multiplying them through (rollout), yielding a class-specific attention map ⁴⁴. These techniques are analogous in spirit to Grad-CAM (which uses gradients to weight feature maps) but applied to the attention matrices – thereby highlighting which patches are attended to for a given class. They tend to produce more focused maps than raw attention. In fact, the chest X-ray study by Kool et al. (2023) used such a gradient-weighted attention rollout (TMME) and found it improved interpretability over vanilla Grad-CAM for ViTs ⁴³ ⁴⁵. The resulting attention-based saliency maps often look cleaner and more class-relevant, since irrelevant attention patterns are down-weighted by the gradient.

Transformer-Specific Attribution (Gradient-based): Another line of SOTA methods adapts the gradient and decomposition ideas from CNNs to transformers. Chefer et al. (2021) introduced a technique often called **Transformer Attribution** (or “Gradient * Relevance” in some references ⁴⁶), which extends the LRP/Deep Taylor Decomposition approach to ViT architectures ⁴⁷ ⁴⁸. Chefer’s method propagates relevance scores through both self-attention layers and the intervening feed-forward layers, carefully handling the unique operations like the softmax in attention and residual connections ⁴⁸ ⁴⁹. The key challenge is to maintain **conservation of relevance** (so that the sum of attributions remains equal at each layer) despite the complex attention mixing. Chefer’s solution was to derive specific propagation rules for attention and for skip-connections (introducing normalization terms to handle the fact that transformers have both positive and negative attributions due to non-ReLU activations) ⁵⁰ ⁵¹. The result is a relevance map over input patches/pixels for the target class, analogous to how LRP works for CNNs. Empirical evaluations showed this method gave more precise explanations than simply using attention weights or naive rollout, especially in multi-object scenes ⁵² ²². Another powerful method is **AttnLRP (Attention-aware LRP)** by Achtibat et al. (ICML 2024), which further refines the LRP approach for transformers. **AttnLRP** introduced improved attribution rules for the key-query-value attention mechanism (hence “attention-aware”) and other transformer components, achieving faithful and holistic attributions with computational cost similar to a single backward pass ⁵³ ⁵⁴. In other words, AttnLRP can propagate relevance not only to the input patches but also through all latent tokens and layers, giving a complete picture of how information flows in the model ⁵⁵ ⁵⁶. Studies have found AttnLRP to be **highly faithful** to model predictions (better preserving the true decision rationale) ⁵⁷. However, one interesting observation is that AttnLRP’s raw maps, while accurate, can appear fragmented or “checkerboard-like” because they assign separate relevance to each patch token ⁵⁸. This can make them harder for humans to parse (less smooth), which has spurred research into combining the best of both worlds: faithfulness of gradient/LRP methods and plausibility (visual coherence) of attention maps.

Hybrid and Improved ViT Explanations: The latest SOTA methods often combine attention with learned filtering or aggregation to yield explanations that are both reliable and human-friendly. For example, Ayyar et al. (2025) in “There is More to Attention” propose **RFEM (Relevance-Filtered Attention Maps)**, which statistically filters the raw multi-head attention maps to remove noise, and optionally introduces class-specific weighting, producing sharper and more interpretable maps ⁵⁹ ⁶⁰. They argue that attention is indeed a valuable signal if processed properly, contrary to the blanket claim “attention is not explanation” ⁶¹. By evaluating on datasets with ground-truth human eye fixations, they showed that their filtered attention maps align better with human perception than previous methods, while still passing faithfulness tests (like perturbation insertion/removal metrics) ⁶² ⁵⁷. This reflects a trend in the field: moving beyond treating raw attention weights as explanations, towards **enhanced attention-**

based explanations that incorporate gradients, statistical priors, or multi-layer context to improve quality ⁶³ ⁶⁴ . Another noteworthy approach is **Score-CAM adaptations for ViT** (e.g., **CDAM: Class-Discriminative Attention Map**, 2022), which avoid using gradients at all – instead, they feed forward masked versions of the image (masking one patch at a time) and use the change in output score to gauge that patch’ s importance, akin to a perturbation approach but aligned with transformer patches ⁶⁵ ⁶⁶ . Such methods can be very faithful but share the high computational cost of perturbation techniques.

In summary, for CNNs, the standard practice for saliency is to apply Grad-CAM or related gradient localization methods ¹⁸ , optionally refined with guided backprop or integrated gradients for detail. For ViTs and attention-based models, practitioners have a choice: one can **visualize raw attention weights** to get a quick sense of model focus, but for rigorous explanations it’ s now common to use **transformer-specific attribution methods** (like attention rollout + gradients, Chefer’ s method, or AttnLRP) that account for the multi-layer attention structure and provide class-specific relevancy ⁴⁶ ⁶⁷ . These SOTA methods represent the current stage of progress: ensuring that even very large, complex models like transformers can be made interpretable through effective saliency mapping.

What Do Visual Attention Maps Reveal?

Saliency and attention maps serve as a window into the functioning of vision models, and over the years they have advanced our understanding of these models’ behavior. We discuss some key insights these maps provide:

- **Model Focus and Feature Importance:** At a basic level, a saliency map shows which parts of the input the model considers important for its prediction. This often confirms that the model is focusing on the intended object or region. For example, a Grad-CAM map for a dog image typically lights up the dog’ s head and body ⁶⁸ , indicating those regions drive the “dog” classification (and not, say, the grass or background). If a map highlights irrelevant regions (like background artifacts), it can reveal that the model may be relying on spurious correlations. This kind of analysis has helped uncover failure cases – a famous anecdote tells of a neural network that learned to detect “camouflaged tanks” by actually looking at clouds in the sky of the photos ⁶⁹ ⁷⁰ . A Grad-CAM visualization would have exposed this, showing heat on the sky instead of the ground where tanks are ⁷¹ ⁷² . Thus, saliency maps act as a **debugging tool**, ensuring the model’ s attention aligns with human expectations and the true signal in the data ⁷³ .
- **Localization and Spatial Context:** Especially in CNNs, class activation maps can serve as rough object localizers. Even though the models are trained for classification, Grad-CAM often localizes the object in the image, sometimes well enough to draw a bounding box around it. This reveals that **CNNs implicitly learn to locate objects** as part of recognizing them. In tasks like weakly-supervised object localization, such maps are directly used to infer object regions. For transformers, attention maps across layers show a progression: early layers might attend broadly or to low-level features (edges, textures), whereas later layers attend more to semantic regions (the object as a whole) ⁷⁴ ⁷⁵ . One study found that in a ViT, by around layer 7 the class token’ s attention map had essentially segmented the object of interest ⁴⁰ . Some patches then lost and regained information in subsequent layers due to residual connections ⁷⁶ , but by the final layer the important object was clearly attended to. This layered attention behavior aligns with the idea that ViTs first gather global information (“feature search”) and later refine it (“conjunction search” combining features) ⁷⁴ ⁷⁷ . In fact, the abstract by Djilali et al. (BMVC 2023) noted that ViTs seem to perform a process analogous to human Feature Integration Theory – first globally scanning for salient features, then integrating them in higher layers ⁷⁷ . Thus, visualizing

attention across layers provides insight into how the model builds its understanding, from edges to objects.

- **Class Discrimination vs. Common Features:** By using class-specific attribution (e.g. Grad-CAM for class A vs class B on the same image), we can see what features distinguish one class from another. For instance, in an image containing both a cat and a dog, an explanation for the “cat” class might highlight the cat, whereas for “dog” it highlights the dog ⁷⁸ ⁷⁹. Sometimes, overlapping heatmaps can reveal interesting biases: if the model always uses one object as context for recognizing another, it might highlight both. Attention maps in multi-object scenes can show whether a model’s prediction for one object is influenced by nearby objects (context). Recent analyses also find that different attention heads in a ViT specialize in different relations – e.g., some heads focus on local texture patterns, others on long-range dependencies or background context ⁸⁰. By examining these, researchers have gained a layer- and head-level understanding of what the model attends to: one head might consistently attend to corners or high-frequency details (like text), another might attend to the center of objects, etc. This diversity of attention is a hallmark of transformer models, as opposed to the more uniform receptive fields in CNNs ²¹ ⁸⁰. It suggests that interpretability might sometimes require aggregating multiple maps or selecting the most relevant head/layer for a given explanation.
- **Faithfulness vs. Plausibility:** There is an ongoing discussion in XAI about whether these visual explanations truly reflect the model’s decision-making (faithfulness) or just provide a human-friendly story (plausibility) ²⁰ ⁶³. Attention weights came under scrutiny when Jain & Wallace (2019) showed in NLP that you can often shuffle attention weights without changing the output – suggesting they are not a causal explanation of the model’s prediction ⁶¹. In response, others (Wiegrefe & Pinter 2019) argued that even if attention is not the sole explanation, it can still be a useful explanation of model behavior ⁸¹. In vision, this debate translates to questions like: if a saliency map is very human-plausible (e.g., highlighting the whole dog), is it definitely faithful? Not necessarily – it might be overstating the model’s true reliance if the model really only needed the dog’s head. On the flip side, a perfectly faithful method might produce a scattered mask that humans struggle to interpret. Current understanding is that **a good explanation should aim for both**: it should not mislead (faithful) but also be intuitively meaningful (plausible) ⁸² ⁸³. This has led to metrics to evaluate explanations: quantitative metrics like deletion/insertion curves (how prediction probability drops when most-salient pixels are removed ¹⁹), and qualitative or human-grounded metrics like how well the map overlaps with human-annotated important regions or human gaze (more on this next) ⁶³ ⁸⁴. The field acknowledges that attention/saliency maps are extremely useful tools, but they should be interpreted with some caution: they highlight correlation with the output, not necessarily a proven causal link (especially for gradient-based methods). Nevertheless, when multiple explanation methods agree or when maps pass sanity checks (e.g., randomizing model weights yields random maps ⁸⁵), we gain confidence that we truly captured something about the model’s inner workings.

In practical terms, visual attention maps have proven their worth in tasks like model debugging, validating robustness, and even in **scientific discovery** (using neural networks to identify what image features correlate with outcomes). For example, in medical imaging, saliency maps can point radiologists to areas the model found abnormal, providing a second set of “eyes” on the image. In autonomous driving, highlighting which regions (road, pedestrians, signs) the model attends to can increase transparency and safety ⁸⁶ ⁸⁷. These maps reveal functionally critical features and offer a form of visual rationale for each prediction, which is invaluable for interpretability.

Impact of Model Size and Efficiency on Visual Attention

As models grow in size and complexity, how do attention and saliency map methods hold up? We consider two aspects: (1) the pattern of attention in larger vs smaller models, and (2) the computational efficiency of extracting saliency from large models.

Attention Patterns vs Model Size: Larger models (e.g. deeper networks or higher capacity transformers) often have more numerous and diverse attention heads or feature channels, which can yield more nuanced attention distributions. Empirically, a bigger CNN (say a ResNet-101 vs ResNet-18) might have finer-grained localization in Grad-CAM – partly because deeper layers can capture higher-level concepts that align well with object parts. However, larger models also sometimes exhibit more diffuse attention: they have the capacity to consider more context. For instance, a small model might focus very tightly on the most discriminative part of an object (like just the face of an animal), whereas a large model might also attend to secondary parts (like the animal’s torso or even its shadow) because it can afford to integrate more information. In ViTs, **increasing the number of heads or layers** allows the model to attend to multiple aspects of the image in parallel – some heads might focus on global context while others on local details. This means the overall attention map (averaged over heads) of a larger ViT could be more spread out across the object and its context, compared to a smaller ViT which might rely on just a few salient patches. Recent analyses (Lopardo et al. 2024) indeed show that different heads in large transformers pick up distinct but complementary patterns ⁸⁰. Thus, with model size, the dimensionality of attention increases – making it even more important to use proper **aggregation or filtering** when producing a single saliency map. SOTA methods like AttnLRP and RFEM were tested on base vs large ViT models and generally scale well, but one observation was that methods assigning per-token scores (like AttnLRP) can produce a “checkerboard” look especially for higher resolution inputs ⁵⁸. This is because each patch is evaluated independently, and with many patches the map can look blocky. Techniques like upsampling and smoothing are used to counteract this and produce a visually coherent result even for high-resolution, large models ²². In summary, larger models can yield richer attention maps, but they also require careful handling to interpret – one may need to decide which layer or head is most relevant to visualize, or to merge many signals. The reward is that large models, when interpreted correctly, might align even better with human-like attention on complex scenes, since they consider more context.

Computational Efficiency Considerations: The cost of generating saliency maps can grow with model size, but recent methods strive to keep it manageable. For gradient-based methods (Grad-CAM, Integrated Gradients, AttnLRP, etc.), the complexity is roughly proportional to doing one (or a few) backward passes through the model. This is usually feasible even for very large models – e.g., AttnLRP’s authors emphasize it has efficiency similar to a single backward pass despite attributing through the whole transformer ⁸⁸ ⁸⁹. By contrast, naive attention-based methods require multiplying $N \times N$ matrices (where N is number of patches/tokens); this is $O(N^2)$, which can be heavy if you try to do full attention rollout on a high-resolution image that was split into many patches ⁹⁰. For example, an image of size 448×448 with 16×16 patches has $N=784$ patches; an 784×784 attention matrix multiplied across 12 layers is not trivial. Methods like **Rollout**, **SAW**, **RFEM** that operate on attention matrices indeed have complexity that grows quadratically with the number of patches ⁹⁰. In practice, though, many ViTs use 224×224 images (196 patches) for which this is fine. Still, if one wanted to get pixel-level resolution by treating each pixel as a token (as some research endeavors in transformer variants), direct attention map multiplication would be infeasible. In such cases, one would revert to gradient or decomposition approaches, whose cost scales with model parameters but not quadratically with input size. A concrete comparison from Ayyar et al. (2025) shows that methods requiring only a backward pass (like Grad-CAM adapted to ViT, or AttnLRP, or Chefer’s grad-based method) tend to be faster than those doing heavy matrix ops on attentions ⁹¹ ⁹⁰. For instance, **Grad-CAM vs. Rollout:**

Grad-CAM (for ViT, meaning taking gradients w.r.t patch embeddings) was found to be among the fastest, whereas pure rollout was very fast too (just forward passes), but rollout lacks class specificity and detail ⁹² ⁹³ . **SAW and class-specific RFEM** – which involve computing gradients of attention – were a bit slower due to that extra backward pass, but still acceptable ⁹¹ . The overhead of even the more advanced methods is often only a small multiple of normal inference time. For example, Integrated Gradients might require 50 forward passes (if 50 interpolation steps) – this is a fixed overhead that doesn’ t explode with model size, though for a huge model those 50 passes might be slow. Perturbation methods like RISE are the ones that suffer most as model and input size grow, since they need many forward passes and each pass on a bigger model is slower. Therefore, **for very large models, practitioners often prefer one-pass gradient methods (Grad-CAM/AttnLRP)** over brute-force perturbation, to keep explanation generation tractable.

There’ s also an interplay between model architecture and ease of interpretability. CNNs have the advantage that their spatial structure is straightforward – one can take the last conv layer and it’ s a natural basis for a heatmap. As CNNs get deeper, you might lose some spatial resolution, but methods like layer aggregation (averaging CAMs from multiple layers ⁹⁴) have been proposed to address that. Transformers, on the other hand, have attention spread across many layers and heads; the challenge is more on the aggregation side than raw compute. In practice, one might use heuristics: e.g., visualize the last layer’ s average attention or a weighted average of a few top layers, if doing it manually. Automated methods like AttnLRP handle the whole stack via backprop, which is efficient and holistic.

Finally, an interesting note: sometimes **larger models yield more stable saliency maps**. They may be less sensitive to tiny perturbations because they have redundant representations, so the saliency methods might produce smoother results. Conversely, a small model at the brink of capacity might have very concentrated, brittle attributions (focusing on one odd feature). This isn’ t a hard rule, but it has been observed in some settings that explanation quality (in terms of highlighting complete objects) can improve with model robustness and capacity ⁹⁵ ⁹⁶ – likely because the model’ s concept of the object is more holistic. Hence, as model size increases, while it’ s computationally heavier to obtain explanations, those explanations may actually become more human-aligned, provided we use the right tools to extract them.

Comparing Model Visual Attention to Human Visual Attention

A compelling question in interpretability is: Do AI vision models attend to images in the same way humans do? Humans have a well-studied pattern of visual attention – we use a foveal gaze, make saccades to salient or task-relevant regions, and exhibit both bottom-up saliency (bright colors, motion, faces attract us) and top-down attention (we look for specific objects when task requires). Saliency maps from models are not exactly eye-tracking heatmaps, but drawing comparisons has been illuminating for researchers.

Similarities: Interestingly, some recent works suggest that modern architectures, especially Transformers, naturally learn attention patterns that correlate with human gaze. “Vision Transformers Are Inherently Saliency Learners” (Djilali et al. 2023) explicitly analyzed ViT attention maps on standard saliency prediction datasets (which have ground-truth human fixation densities) ⁷⁴ ⁷⁵ . They found that with minor smoothing or processing, the ViT’ s attention maps achieved reasonable scores in predicting human saliency ⁷⁵ . The ViT’ s multi-head attention seems to first broadly attend to all instances of certain features (like a human scanning for anything salient) and then concentrate on important combinations of features, reminiscent of human visual search ⁹⁷ . They draw analogies to the **Feature Integration Theory** in cognitive science – which posits that humans first do parallel feature detection, then a focused attention for feature conjunctions ⁹⁷ . The ViT appears to mimic this: early layers have

heads that globally highlight salient edges or textures, later layers have heads that combine those into object-level attention ⁷⁷. This alignment suggests that at a coarse level, model attention (especially in architectures designed with global self-attention) can be interpretable through the lens of human attention mechanisms. Even CNN saliency maps often end up highlighting objects or regions that a human might also find important for a task (e.g., when asked “is there a cat?”, both human and model would focus on the cat). In fields like medical imaging, where radiologists’ eye fixations are recorded, researchers have compared model saliency to expert gaze. Often, a well-trained model’s Grad-CAM or attention map overlaps significantly with the regions doctors examine, such as a specific abnormality on an X-ray ^{98 99}. This is encouraging: it means the model is “looking” where an expert would look, hinting at a form of common reasoning or at least that the model isn’t completely alien in how it processes the image.

Differences: Despite these overlaps, there are notable differences between model and human attention. Humans have **sequential attention** – we move our gaze and can only focus on a few degrees of vision at once. Models (especially ViTs) attend globally in parallel; their “attention map” is not constrained by a single foveal spot. This means a model might assign fractional attention to many regions simultaneously, whereas a human would serially inspect them. For example, a transformer may have one head attending to a background element while another attends to the foreground object at the same time ⁸⁰, reflecting a multi-focal attention that humans lack. Additionally, human attention is driven by semantics and curiosity – we tend to look at faces, text, or novel anomalies in a scene even if not necessary for the task. Models, unless explicitly trained to do so, will only attend to what maximizes task performance. This can lead to cases where a model’s saliency map and human gaze differ: e.g., in an image classification, a human might glance at the entire animal, but a CNN might zero in only on the distinctive fur pattern on a flank because that’s enough to classify. Conversely, a model might highlight a subtle texture that humans overlook. One study on object detection found that model “attention” (via gradients) sometimes strongly weights regions that humans wouldn’t think of as salient – for instance, the model might use a shadow to infer object presence, whereas humans focus on the object silhouette ¹⁰⁰.

Crucially, human attention is often taken as a **benchmark for explanation plausibility**. If an explanation map aligns well with where humans look to identify the object, we consider it more intuitive. Researchers have started to use eye-tracking datasets to quantitatively evaluate this alignment: metrics like Similarity (SIM), Pearson correlation (PCC) between the explanation heatmap and human fixation heatmap, AUC-Judd, etc., measure how close the model’s attention is to human attention ^{101 84}. In Ayyar et al. (2025), for example, they report these metrics and show that their RFEM method achieved higher correlation with human gaze than prior methods like raw rollout or AttnLRP ^{102 57}. Interestingly, they note that AttnLRP, while very faithful, produced a more checkered map that **did not match human attention as well** (lower plausibility) ⁵⁸. Their filtered attention maps, on the other hand, were both faithful and more human-aligned ⁵⁷. This underscores that while human attention isn’t the ground-truth of model correctness, it’s a valuable reference point: if an AI’s “attention” aligns with human common sense, users feel more comfortable with it.

Another facet is using human attention to **guide or improve models**. A novel direction is “gaze-informed neural networks.” For instance, Koorathota et al. (2023) (“Fixating on Attention: Integrating Human Eye Tracking into Vision Transformers”) took human driver gaze data and imposed a loss so that the ViT’s attention weights align with where human drivers look in critical driving scenes ^{103 104}. They introduced a fixation-attention intersection (FAX) loss to encourage overlap between model attention and human fixation maps ¹⁰⁵. The result was not only a model that had attention maps more similar to human gaze, but also one that **performed better** under uncertain conditions (it made more accurate driving decisions) ^{106 104}. This suggests a synergistic relationship: human attention can be used as an additional supervisory signal to shape model attention in a beneficial way, effectively embedding human priors about what is salient. Meanwhile, the fact that doing so improved performance hints that the

human way of perceiving scenes has efficiencies that the model can benefit from – perhaps focusing on truly relevant areas improves generalization.

Human vs Model Attention in Practice: In practical interpretability studies, one might overlay a model’s saliency map with human eye-tracking heatmaps for the same image. Often there is partial overlap – e.g., both highlight the main object – but humans might also look at secondary objects out of curiosity, whereas the model might highlight only the label-relevant parts. For instance, if an image shows a dog and a ball, a human might look at both dog and ball (especially if they anticipate interaction), but a classifier labeling “dog” will ignore the ball and only highlight the dog. Neither is “wrong” – they have different goals. Understanding these differences is important when designing explanations: an explanation that purely matches human attention may omit some model-specific quirks (like reliance on background). That’s why the field distinguishes plausibility (looks convincing to a human) from faithfulness (accurately reflects the model) ⁶³. The best-case scenario is when these coincide – many vision models, when they work well, do end up attending to what a person would intuitively think of (e.g., focusing on a tumor in a scan, or the road ahead in driving). But when they diverge, those divergences are informative: they either highlight a model flaw or perhaps an unexpected strategy the model found.

In conclusion, model visual attention maps have a meaningful but not one-to-one relationship with human visual attention. Current research often uses human attention as a yardstick to evaluate and improve model explanations ^{98 99}. The convergence of evidence (like ViT attention mirroring human saliency in some cases ^{74 107}) is an exciting development – it not only validates that these models “see” in a recognizable way, but also opens the door to human-AI collaborative vision. By understanding and aligning model attention with human attention, we can build AI systems that are more transparent and perhaps even teach models to attend better (and vice versa, use models to discover where humans should look, as decision support).

Conclusion

The field of visual interpretability has evolved from simple gradient maps on early CNNs to sophisticated attention-based explanations on today’s state-of-the-art architectures. **Standard methods** like Grad-CAM, integrated gradients, and occlusion analysis remain widely used due to their simplicity and effectiveness in highlighting important image regions ^{2 11}. At the same time, **cutting-edge techniques** tailor the approach to the model architecture: for Vision Transformers and other attention networks, we now leverage the internal attention structure (with methods such as attention rollout and AttnLRP) to obtain explanations that are both finer and more faithful ^{60 89}. The current stage of progress emphasizes a dual objective: **faithfulness** (the explanation truly reflects the model’s decision logic) and **human interpretability** (the explanation is understandable and convincing to people) ^{108 63}. Achieving both is an active area of research, as seen by methods that refine raw model attention into human-aligned saliency maps without sacrificing correctness ^{57 58}.

From a practical standpoint, if one is undertaking a project to extract and use AI model attention/saliency (as you intend for your project’s methodology), here are some guidelines informed by the literature:

- For **CNN models**, a recommended starting point is **Grad-CAM or Grad-CAM++** for quick visual explanations ¹¹. These provide class-specific heatmaps with a single backward pass, and many libraries (e.g. PyTorch’s Captum, TensorFlow’s explainability toolkit) have implementations. If higher resolution detail is needed, combine Grad-CAM with guided backprop (Guided Grad-CAM) to visualize fine features on top of the coarse localization ^{14 15}. For rigorous analysis, consider also **Integrated Gradients** to check consistency of attributions, and a perturbation test (like

remove top-10% salient pixels and see if confidence drops significantly) to validate the saliency map’ s relevance ¹⁹ .

- For **Vision Transformer models**, try to exploit their attention mechanism: visualize the raw attention of the last layer to get an initial sense of model focus, but for a true explanation use a method like **Attention Rollout with class gradients (e.g. SAW)** or **AttnLRP** ⁶⁷ ⁸⁹ . Libraries and code exist (e.g. the authors of these methods often release code – Chefer’ s TransformerExplainability code, or the LRP for Transformers library from Achibat et al.). These will give you patch-level importances. You may want to overlay or upsample those to pixel level for presentation. Keep in mind that ViT explanations might need smoothing – simple techniques like bilinear interpolation of patch scores or applying a Gaussian filter can make the visualization more continuous (as done by some authors to mimic human-like saliency maps ⁷⁵). Always verify that the explanation is pointing to the expected object or region (if it’ s wildly highlighting something odd, the model might be picking up on spurious correlations).
- Be mindful of **efficiency vs. completeness**: if your project requires processing thousands of images for saliency, you might avoid very slow methods like exhaustive perturbations. Instead, favor one-pass methods (Grad-CAM, rollout). If you need extremely fine-grained explanations (pixel-perfect segmentation-level), note that these techniques are not exact segmentation tools – they provide heatmaps, which are usually sufficient to indicate important regions but not to precisely outline objects. For finer results, sometimes people combine saliency with segmentation algorithms (e.g., using the saliency map as a prior for a segmentation model). Also, as models get larger, trust that these explanation methods have been tested and generally scale; just ensure you have the computational resources, especially if using gradients on a huge transformer – memory can be a concern for full gradient propagation through all layers.
- Lastly, incorporate **human perspective** appropriately: if your domain has human attention data or ground-truth important regions, compare the model’ s saliency maps to those. Discrepancies might highlight model errors or biases that you can address. Conversely, if the model consistently attends to something humans ignore, investigate if that feature is legitimately predictive or a flaw. The end goal is not just to generate saliency maps, but to use them to interpret and improve your model. As the literature shows, attention maps can foster trust and catch failure modes, but they should be used with an understanding of their limitations.

In the broader context, this review underlines that **attention and saliency extraction for visual models has become a mature but still rapidly advancing field**. We now have a spectrum of tools from simple to sophisticated, each with trade-offs, and a much better grasp of what these visual explanations mean. Moreover, comparisons to human attention provide a fascinating convergence between machine perception and human cognition ⁷⁷ ¹⁰⁵ . By applying these SOTA practices in your project, you’ ll be able to dissect your AI model’ s “visual thought process” and ensure that its attention aligns with both the task objectives and, ideally, human common sense – a key step towards transparent and reliable AI in computer vision.

Sources: Selvaraju et al. 2017; Chattopadhyay et al. 2018; Sundararajan et al. 2017; Bach et al. 2015; Abnar & Zuidema 2020; Chefer et al. 2021; Achibat et al. 2024 (AttnLRP); Mallick et al. 2022 (SAW); Ayyar et al. 2025 (RFEM); Djilali et al. 2023; Koorathota et al. 2023; and other works ² ⁶⁰ ⁶⁷ ⁷⁴ ¹⁰⁶ , as cited throughout.

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Explanation maps of (a) SOTA AttnLRP (b) Gradient $\times$ Relevance, (d) GradCAM, (e) Input $\times$ Gradient, (f) Attention Rollout, (g) SAW and our methods (h) RFEM and the class-specific (i) RFEM-Class

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